

Antisymmetric eigenfunctions disprove separation for the pseudo- p -Laplacian

A counterexample to the question in arXiv:2006.07948

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Status. Candidate counterexample, likely valid, pending human review. For every $1 < p < \infty$, the Dirichlet pseudo- p -Laplacian on the square has a nonzero weak eigenfunction which is antisymmetric across the diagonal. It cannot be a separated product of one-dimensional p -sines. Thus the literal classification asked in the source is false, in particular when $p \neq 2$.

1 Source question

For a rectangle $R = \prod_{i=1}^n (a_i, b_i)$, Edmunds, Lang, and Mihula define

$$\tilde{\Delta}_p u = \sum_{i=1}^n \frac{\partial}{\partial x_i} \left(\left| \frac{\partial u}{\partial x_i} \right|^{p-2} \frac{\partial u}{\partial x_i} \right).$$

After identifying its first Dirichlet eigenfunction, Remark 3.8 asks whether *all* eigenfunctions are products

$$\prod_{i=1}^n \sin_p \left(\frac{\pi_p k_i (x_i - a_i)}{b_i - a_i} \right), \quad k_i \in \mathbb{N},$$

and records this as open for $p \neq 2$ [1].

$$\frac{u \in W_0^{1,p}(D)}{u \neq 0} \quad \| \cdot \|_{L^p(R)}$$

was proved in [2, Lemma 2.1]. $\square$

Remark 3.8. It can be routinely shown that the extreme function for the Rayleigh quotient

$$\inf_{\substack{u \in W_0^{1,p}(D) \\ u \neq 0}} \frac{\| |\nabla u| \|_{L^p(R)}^p}{\| u \|_{L^p(R)}^p}$$

is the first eigenvalue of the pseudo- p -Laplacian operator with Dirichlet boundary conditions, i.e.:

$$(3.18) \quad \tilde{\Delta}_p u = \tilde{\lambda}_p |u|^{p-2} u, \quad \text{with } u = 0 \text{ on } \partial R,$$

where

$$\tilde{\Delta}_p u = \sum_{i=1}^n \frac{\partial}{\partial x_i} \left(\left| \frac{\partial u}{\partial x_i} \right|^{p-2} \frac{\partial u}{\partial x_i} \right).$$

Then it follows from Corollary 3.7 that the first eigenfunction for the Dirichlet problem (3.18) for the pseudo- p -Laplacian operator on the domain R is the function defined by (3.17). In the case where R is a cube, this was already observed in [2, Example 2.4].

This coincides well with the classical result for $p = 2$. However, whether all eigenfunctions for (3.18) on the domain R are of the form

$$\prod_{i=1}^n \sin_p \left(\frac{\pi_p k_i (x_i - a_i)}{b_i - a_i} \right), \quad (x_1, \dots, x_n) \in R, \text{ for some } k_i \in \mathbb{N},$$

remains an open question if $p \neq 2$.

Not only is the canonical embedding $W_0^{1,p}(D) \hookrightarrow L^p(D)$ maximally noncompact, but also all its strict s -numbers coincide with the norm of the embedding.

Theorem 3.9. Let $p \in (1, \infty)$. Let $k \in \{1, \dots, n-1\}$ and $-\infty < a_i < b_i < \infty$, $i = 1, \dots, n-k$.

Set $D = \mathbb{R}^k \times (a_1, b_1) \times \prod_{i=1}^{n-k} (a_i, b_i) \subseteq \mathbb{R}^n$. We have that

$$a_m(I) = b_m(I) = c_m(I) = d_m(I) = i_m(I) = m_n(I)$$

We use the standard positive-eigenvalue weak convention

$$\sum_{i=1}^n \int_R |u_{x_i}|^{p-2} u_{x_i} \varphi_{x_i} = \lambda \int_R |u|^{p-2} u \varphi, \quad \varphi \in W_0^{1,p}(R). \quad (1)$$

Thus the source's displayed strong equation has $\tilde{\lambda}_p = -\lambda$.

2 Proof intuition

On the square, combine transposition of the two coordinates with a minus sign:

$$(Su)(x, y) = -u(y, x).$$

Both the L^p mass and the anisotropic p -energy are invariant under this involution. Minimize the Rayleigh quotient among its fixed points, which are exactly the antisymmetric functions. Compactness gives a minimizer. An elementary averaging argument shows that a constrained critical point in the fixed space is a critical point against *all* variations, so it is a genuine eigenfunction. Antisymmetry is incompatible with any nonzero separated product: on the diagonal it would force the product of two p -sines to vanish identically.

3 The counterexample

Theorem 1. Let $1 < p < \infty$ and $Q = (0, 1)^2$. There exist $\lambda > 0$ and a nonzero $u \in W_0^{1,p}(Q)$ satisfying (1) and

$$u(x, y) = -u(y, x) \quad \text{for almost every } (x, y) \in Q.$$

Consequently u is not a nonzero scalar multiple of

$$\sin_p(\pi_p k x) \sin_p(\pi_p \ell y) \quad (k, \ell \in \mathbb{N}).$$

In particular, the separated-product classification in Remark 3.8 of the source is false for every $p \neq 2$.

Proof. Put $X = W_0^{1,p}(Q)$ and define

$$E(v) = \int_Q (|v_x|^p + |v_y|^p), \quad N(v) = \int_Q |v|^p.$$

The map $S : X \rightarrow X$, $(Sv)(x, y) = -v(y, x)$, is a linear isometric involution, and $E(Sv) = E(v)$ and $N(Sv) = N(v)$. Its fixed space

$$X^- := \{v \in X : Sv = v\} = \{v \in X : v(x, y) = -v(y, x) \text{ a.e.}\}$$

is closed and nonzero. Indeed, if $\eta, \zeta \in C_c^\infty(0, 1)$ are linearly independent, then $\eta(x)\zeta(y) - \eta(y)\zeta(x)$ is a nonzero member of X^- .

Minimize E over

$$\mathcal{M}^- := \{v \in X^- : N(v) = 1\}.$$

A minimizing sequence is bounded in $W_0^{1,p}(Q)$ by Poincaré's inequality. Reflexivity, weak closedness of X^- , compactness of $W_0^{1,p}(Q) \hookrightarrow L^p(Q)$, and weak lower semicontinuity of E give a minimizer $u \in \mathcal{M}^-$. The constraint is regular since $N'(u)[u] = p$. The Lagrange multiplier theorem on X^- therefore gives a $\lambda \in \mathbb{R}$ such that

$$J'(u)[w] = 0 \quad (w \in X^-), \quad J := E - \lambda N.$$

Testing with $w = u$ yields $\lambda = E(u) > 0$ because $N(u) = 1$.

It remains to justify that no variations were lost by restricting to X^- . The functional J is S -invariant and $Su = u$, hence differentiating $J(S(u + tv)) = J(u + tv)$ at $t = 0$ gives

$$J'(u)[Sv] = J'(u)[v] \quad (v \in X).$$

For an arbitrary $v \in X$, let $Pv = (v + Sv)/2 \in X^-$. Then

$$J'(u)[v] = J'(u)[Pv] = 0.$$

Writing out this identity and cancelling the common factor p gives (1) for every test function $\varphi \in X$. Thus u is a genuine weak Dirichlet eigenfunction, not merely a critical point within the symmetry class.

Suppose, for a contradiction, that u were a nonzero scalar multiple of a displayed separated product. Combining this equality almost everywhere with $u(x, y) = -u(y, x)$ shows that the continuous function

$$h(x, y) = \sin_p(\pi_p kx) \sin_p(\pi_p \ell y) + \sin_p(\pi_p ky) \sin_p(\pi_p \ell x)$$

vanishes almost everywhere, hence everywhere. On the diagonal this says

$$2 \sin_p(\pi_p kt) \sin_p(\pi_p \ell t) = 0 \quad (0 < t < 1).$$

Each factor has only finitely many zeros in $(0, 1)$, so one can choose t where both are nonzero, a contradiction. $\square$

4 Deep upgrade: all cubes

Theorem 2. *For every $n \geq 2$, $1 < p < \infty$, and cube $Q_n = (0, 1)^n$, the Dirichlet pseudo- p -Laplacian has a nonzero weak eigenfunction satisfying*

$$u(x_1, x_2, x_3, \dots, x_n) = -u(x_2, x_1, x_3, \dots, x_n).$$

It is not a nonzero scalar multiple of any separated p -sine product. The same conclusion holds on any rectangular box with two congruent coordinate intervals, after translating those coordinates before swapping them.

Proof. Use the signed transposition $(Su)(x_1, x_2, x') = -u(x_2, x_1, x')$. The functionals

$$E(u) = \int_{Q_n} \sum_{i=1}^n |u_{x_i}|^p, \quad N(u) = \int_{Q_n} |u|^p$$

remain invariant. The fixed space is nonzero: multiply the two-dimensional test function used above by any nonzero element of $C_c^\infty((0, 1)^{n-2})$. The direct-method and averaging proof of Theorem 1 applies verbatim. A separated product cannot have the signed-swap symmetry: choose the remaining coordinates away from the finite zero sets of their factors and then repeat the diagonal argument in x_1, x_2 . Equal-length translated intervals are handled by the affine coordinate transposition between them. $\square$

5 Novelty, interpretation, and scope

The first attempt was the square construction. The required deep upgrade abstracted the signed-involution mechanism, proved full criticality directly instead of invoking a black-box symmetric-criticality theorem, and extended the counterexample to every cube and to boxes with two equal side lengths.

A bounded primary-source search used the exact source question together with “pseudo- p -Laplacian”, “rectangle”, “eigenfunction”, “antisymmetric”, and “coordinate swap”. It found the source and work on symmetry for the isotropic p -Laplacian or on nonlocal pseudo- p operators, but no primary paper giving this counterexample to the displayed rectangle classification. Novelty confidence is moderate because the variational argument is elementary; correctness confidence is high.

The conclusion concerns the literal claim that every eigenfunction itself is one separated product. It does not classify all eigenvalues, decide whether separated modes account for the spectrum in some weaker sense, or determine the least eigenvalue within the antisymmetric class. The same construction also produces antisymmetric eigenfunctions at $p = 2$, as expected from degenerate eigenspaces on symmetric boxes; the source asks specifically for $p \neq 2$.

References

- [1] D. E. Edmunds, J. Lang, and Z. Mihula, *Measure of noncompactness of Sobolev embeddings on strip-like domains*, arXiv:2006.07948.